Power-board battery case comparison

Assembly fit, bolt tightening, side pull and vibration response

9 November 2026 | Native structural analysis in ANSYS MAPDL 2026 R1.02

The mounting height determines whether the cells are already loaded

Placing the four tabs directly against the board creates approximately 1.18 mm of interference at the cell crowns. Each of the three cell solids shares about 1.70 cm^3 with the undeformed enclosure. A 3 mm tab stand-off clears the cells, nickel strips, washers and fasteners present in the 35-body mechanical STEP assembly. Thus a flush installation must displace or deform parts as the screws tighten. The raised case begins with an air gap. Translating a rigid cell along the PCB X axis first closes a gap after 0.146–0.325 mm, depending on the cell and direction. Flexible relative motion determines whether contact occurs under load.

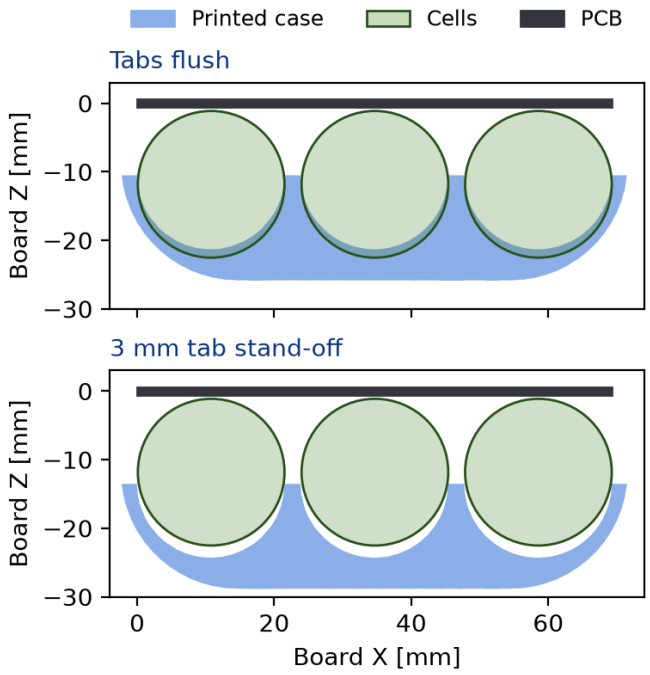


Figure 1: Section through the middle of the cells, in PCB coordinates. The solid outlines show as-designed geometry before tightening. The mounting tabs lie beyond this section.

Reference assembly

The October assembly has 208 bodies and a modeled mass of 251.82 g. Eight stand-off end faces carry the base motion. Two nickel strips attach each cell to the assembly. In the connected reference, six existing screws were loaded to 750 N apiece and locked. Its first mode was 485.964 Hz. Independent base excitation between 20 and 2000 Hz gave a maximum PCB directional displacement of 0.052674 mm RMS under Z excitation and a maximum PCB equivalent stress of 90.325 MPa RMS under X excitation. These archived quantities supply reference checks for the repeats.

Registration and fastener selection

The four nonplated corner holes in PCB1 ODB++ have finished diameter 4.200 mm. The hole pattern measures 62.865×114.427 mm. The case bores have two diameters, 5.000 and 5.090 mm, on a 62.855×113.241 mm pattern. Centring the two patterns leaves about 0.593 mm of offset at each end. The sum of the radial clearances of a nominal 4 mm shank in the smaller case bore and in the board is 0.600 mm. The approximately 0.007 mm difference is the available nominal alignment margin there. Printed hole size, fastener major diameter and drilling tolerances therefore matter for installation.

Tab stand-off [mm]	Cell overlap [mm^3/cell]	Washer bodies intersected	Other intersections
0	1696–1698	4	6 nickel strips
1.2715	0	4	None found
2	0	4	None found
3	0	0	None found

Table 1: Solid intersections between the unstrained case and mechanical STEP bodies. The board's lower face defines the tab seating datum.

Materials and mounting representation

The supplied single-solid STEP has volume 63.069 cm^3. At a printed-PLA reference density of 1.17 g/cm^3, a dense case weighs 73.79 g, about 29.3% of the bare assembly mass. The material axes place the print layers parallel to the PCB for this scenario. PolyLite PLA coupon tensile moduli are 3.427 GPa in the print plane and 3.065 GPa through the layers. Shear moduli and Poisson ratios are engineering assumptions. Actual infill, orientation, print parameters and case mass require measurement. The published coupons used 100% infill. Polymaker technical data sheet.

The candidate model uses four M4 beam shanks with an effective tensile area of 8.78 mm^2. The comparison uses 69 GPa for aluminium and an exploratory 3.0 GPa for nylon. The nominal nylon hardware is McMaster-Carr 93640A145, M4 × 0.7 thread, 12 mm long. The represented bearing patches transfer loads between each end and its solid mesh. Compression-only members represent tab seating, with an assumed PLA annulus at the raised attachment points. The four case screws have no measured installation torque. The 750 N in the reference applies to the six original battery fasteners. McMaster-Carr metric socket-head screws.

Numerical checks and remaining comparisons

The initial mesh has 16,108 curved quadratic tetrahedra. A 43,659-element refinement is available. Both have positive minimum scaled Jacobians. The native preparation check counted four screw beams, four compression seats and 8,038 candidate case contact faces. The contact calculation must establish residual penetration, support reactions, seated shape and stresses before modal or random-response comparisons can support a performance claim. In the October nickel-damage study an assumed material law first registered damage around 43 N during a partial pull. A physical breaking load remains undetermined.

The bare-board repeat reproduces the six locked bolt loads

The original battery attachment provides the initial stress field for the connected modal and vibration analyses. In the first nonlinear load step each of the six pretension elements receives 750 N. In the second step MAPDL holds the attained section adjustment fixed. The reaction at the pretension degree of freedom then measures how much load the locked joint retains. The four case screws are absent from this reference run.

Existing bolt	Locked adjustment [μm]	Locked reaction [N]
1	28.297	749.99935
2	22.205	749.99945
3	18.597	749.99908
4	30.457	749.99680
5	20.686	749.99942
6	18.772	749.99868

Table 2: Fresh MAPDL extraction after length locking. Every row agrees with the October export at its stored precision. The largest departure from the applied 750 N is 0.00321 N.

The solver stored eight equilibrium states and completed the two load steps in approximately 59 minutes on 12 distributed processes. The largest total displacement at lock was 0.0474595 mm. The repeated result differs from the October Mechanical export by 3.4×10^{-8} in relative terms. The model produced 45 warnings, including overlapping contacts, automatically removed redundant constraints and the fixed orientation of pretension elements during large-displacement analysis. The warning text printed in the main solver listing matched the reference run. This comparison establishes repeatability for the existing attachment. Page 7 compares the locked bare and raised assemblies. The vibration response requires separate modal and random-response results.

How the side-pull comparison measures load transfer

The local model drives the outer cell sideways through its rigid pilot while the two nickel strips bend. In the washer-clamped fixture, each strip is compressed with 750 N and the clamp height is then locked. During the pull, the cell can translate in the other two directions and rotate. The cased fixture adds a deformable printed enclosure, four screw shanks and compression-only tab seats. The neighboring cells and screw tops are fixed in this local approximation. The native model check counted 9,502 potential PLA contact faces. These contacts carry compression between PLA and the cell or washer surfaces. Direct PLA-to-nickel contact is omitted from this local fixture. That omission matters most in the flush geometry, where the CAD also intersects the strips.

For each converged load increment the audit sums external forces and moments about the displaced cell centre. It includes the case screw supports and the neighboring-cell support when present. This makes it possible to test whether enclosure contact relieves the nickel strips at the cost of increased screw or board load. The check excluded unconverged diagnostic result sets. On historical friction fixtures it recovered the native force and moment sums across nominal, refined and reverse-direction pulls. The fresh bare nominal fixture completed all 33 substeps in sequence. At 0.500 mm displacement its lateral reaction was 8.542 N and the reported maximum equivalent plastic strain was 0.01254, matching the October fixture. The maximum lateral force imbalance was 2.34×10^{-5} N. This endpoint describes the stiffness of the bilinear nickel fixture under the specified pull. A rupture load requires a failure model and evidence of separation; the raised/nylon result is presented on page 6.

Both default-contact meshes reproduce the October modal spectra

Each unpreloaded model used the archived mounting constraints and searched for 20 undamped natural frequencies between 1 and 10,000 Hz. The original mesh returned modes from 62.805 to 2205.547 Hz and the refined mesh from 66.611 to 2247.595 Hz. The maximum relative differences from the corresponding October exports were 1.78×10^{-8} and 1.74×10^{-9}. Repeating a given discretization thus produced the same frequency spectrum to the stated precision.

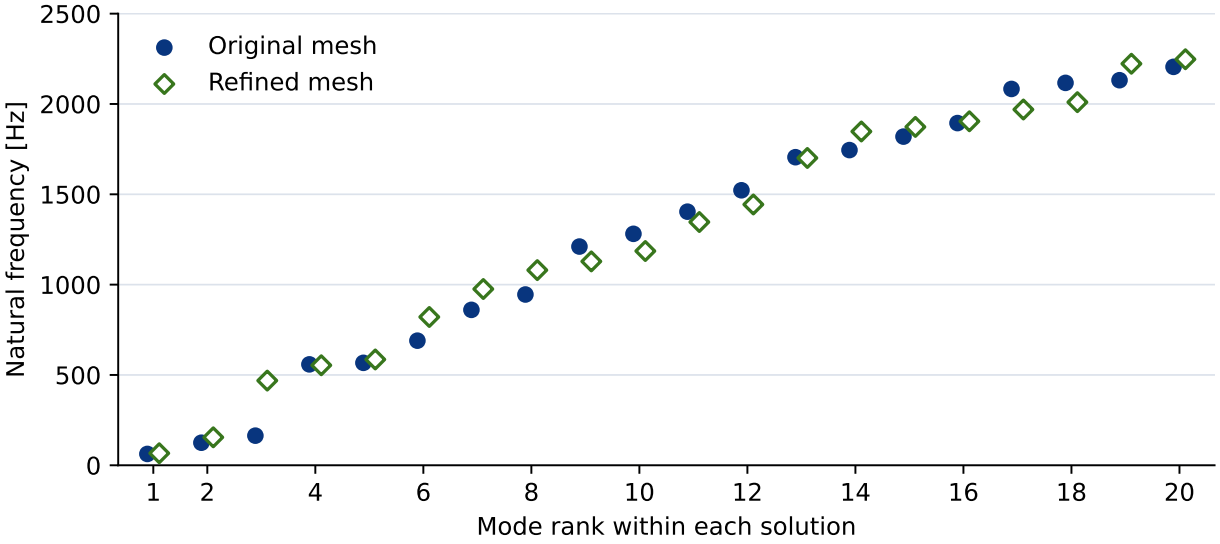


Figure 2: Fresh ANSYS results with the original default contacts. Rank orders the frequencies within each mesh. A shared rank can contain different mode shapes, particularly where modes exchange order.

The first original-mesh run returned only four modes and exited normally. The repeat kept the same input geometry, material properties, supports and search range. It changed the Block Lanczos algorithm to its alternate formulation and requested the Sturm sequence check. This recovered the complete spectrum. The default models retain the detached nine-body group in the earlier connection diagnosis. The bearing, locking and contact states of the six-bolt model provide the starting point for the connected vibration comparison.

How the seeded motion and stress movies will be checked

The three accelerations are separate deterministic realizations of the October 20–2000 Hz base spectrum. Each contains 589,825 samples over 9 s at 65,536 Hz. The input rises over 0.1 s. During the final eight seconds the periodic waveform has 14.136 g RMS. The native transient integration uses the retained prestressed modes, 2% modal damping and motion relative to the common support. Each clip uses 480 signed states from 1.000061 to 1.029297 s, displayed at 60 frames per second.

The PCB stress movie combines the six signed stress components of every mode with the corresponding time-dependent modal coordinate. It calculates von Mises stress separately at every element corner. At an interface, two neighboring elements can return different values. The stress field contains the dynamic fluctuation about the preloaded equilibrium. On the October reference, the independent reconstruction reproduced all 480 native frame maxima within 1.1×10^{-7} Pa across 39,014 PCB elements. The peak for that specific historical X-axis realization was 154.41 MPa. Each new comparison requires its own modal basis, coordinate history and stress recovery.

Does the contact model start with the gap measured in CAD?

An enclosure that presses on the cells at the start of a raised-case calculation will appear to stiffen the assembly before the real parts would touch. We checked the undeformed contact pairs separately from the load solution. The local fixture uses one target for the moving edge cell and another for the fixed neighboring cells and washers. The global assembly uses deformable targets. The search radius tells MAPDL which nearby facets to examine; it does not set the physical clearance.

At 3 mm tab stand-off	Default search	0.6 mm search	2 mm search
Local: moving edge cell	+0.212	+0.212	+0.212
Local: fixed cells and washers	−7.685	+0.200	+0.200
Global: enclosure to assembly	+0.199	+0.199	+0.199

Table 3: Initial native contact check in millimetres. Positive values are gaps; negative values are overlaps. Default search radii ranged from 10.6 to 11.1 mm. The 1 mm search reproduced the 0.6 and 2 mm columns.

The CAD solids do not intersect at this stand-off. We rejected the first raised local calculation because its 7.685 mm overlap contradicted that independent check. It did not reach a converged load increment. The replacement uses a 1 mm search radius for the two added contact pairs and leaves the nickel clamp contacts unchanged. The comparison calls for the cell travel at first closure, contact pressure, penetration, and load transferred to the case mounts.

The finite-element gap is sampled at contact detection points on a curved mesh. The closest point on each exact cell surface is 0.144–0.161 mm from the case. The sideways travel to first rigid contact at the edge cell is 0.163 mm in positive X and 0.308 mm in negative X. Those are different measurements: the shortest distance can be oblique, and a loaded cell can rotate. Agreement among search radii checks the search procedure; it does not establish mesh convergence of contact onset or case stiffness.

Why the flush installation needs a separate check

The flush STEP has real interference. In the local model, a 2 mm search recovered the 1.178 mm crown overlap of the moving cell. A 1 mm search found only 1.000 mm, and a 0.6 mm search found only 0.598 mm. Tightening the search around the flush configuration can therefore hide real interference. Conversely, the default global search reported a peak overlap of 10.093 mm, versus 1.848 mm with a 2 mm search. These values identify a contact-model discrepancy. They do not establish an installation force. The flush comparison requires a verified path into contact, including the washers and strips, and force balance in the installed assembly before the vibration solve.

The raised case restrains the edge cell once its air gap closes

In the first completed cased comparison, the four nylon screws held the printed case at a 3 mm stand-off. As before, a pilot moved the outer cell in positive X while leaving its other translations and rotations free. The two force curves follow one another until the moving cell engages the enclosure. The last open state was at 0.1956 mm and the first closed state at 0.2206 mm. The rigid CAD translation of 0.1634 mm provides a reference for the geometry; compliance, rotation and contact-surface discretization affect the native onset.

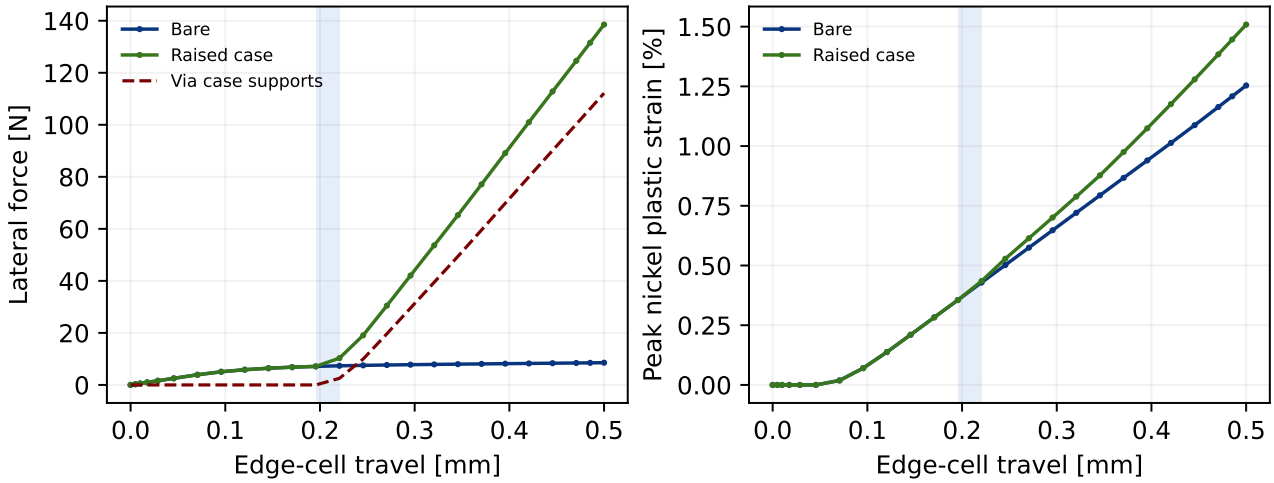


Figure 3: Native converged states in the bare and raised/nylon fixtures. Shading brackets the onset of cell-to-case contact. The dashed line is the force transmitted through the case supports. The right panel shows the largest reported equivalent plastic strain in the two active nickel strips.

At 0.500 mm prescribed cell travel	Bare fixture	Raised case, nylon
Cell reaction in X [N]	8.54	138.53
Load reaching the four case screw supports in X [N]	—	112.15
Load reaching the original clamp supports in X [N]	8.54	26.38
Sideways work applied to the cell [mJ]	3.30	21.41
Peak equivalent plastic strain in nickel [%]	1.254	1.509

At the same travel the cell required 16.2 times as much force with the case. About 81% of that force reached the four screw supports and none reached the fixed neighboring-cell/washer support. At the same *force* as the bare endpoint (8.54 N), the raised fixture instead traveled between 0.196 and 0.221 mm; linear interpolation gives 0.207 mm. At the two cased states bracketing 8.54 N (7.13 and 10.32 N), peak nickel plastic strain was 0.355% and 0.435%, compared with 1.254% at the bare endpoint. Interpolation gives about 0.39%; the solver did not impose 8.54 N on the cased fixture. Forcing both systems to 0.50 mm instead increased the cased peak strain by 20%. Integrating reaction over travel gives 3.30 mJ bare and 21.41 mJ cased. This is external work at the cell; it includes elastic storage and does not measure energy dissipated in the case.

How much confidence to place in the comparison

Both fixtures completed 33 consecutive equilibrium states. At the cased endpoint, 15 active-cell contact faces were closed. Across the full path, the largest penetration at a closed corner was 0.0116 mm. Independently summing the native contact forces agreed with the case-support resultant to within 0.00086 N. The two nickel strips have no direct contact with the raised enclosure in this local model. The fixed screw tops represent a stiffer support than a flexible PCB can provide. These numbers describe one mesh, one pull direction and one mounting assumption. They neither measure nickel rupture nor establish vibration protection. Reverse and refined pulls, flexible-board response and the other screw/seating variants remain open.

The raised nylon-screw case reduces board deflection during tightening

We compared the two flexible assemblies after loading each of the six original bolts to 750 N and locking its attained length. In the cased assembly the four corner screws connected the printed case to the board at a 3 mm stand-off. No base vibration or side pull was applied. Both runs reached all eight equilibrium states. At the final state, each original bolt retained 749.996–750.000 N in both configurations.

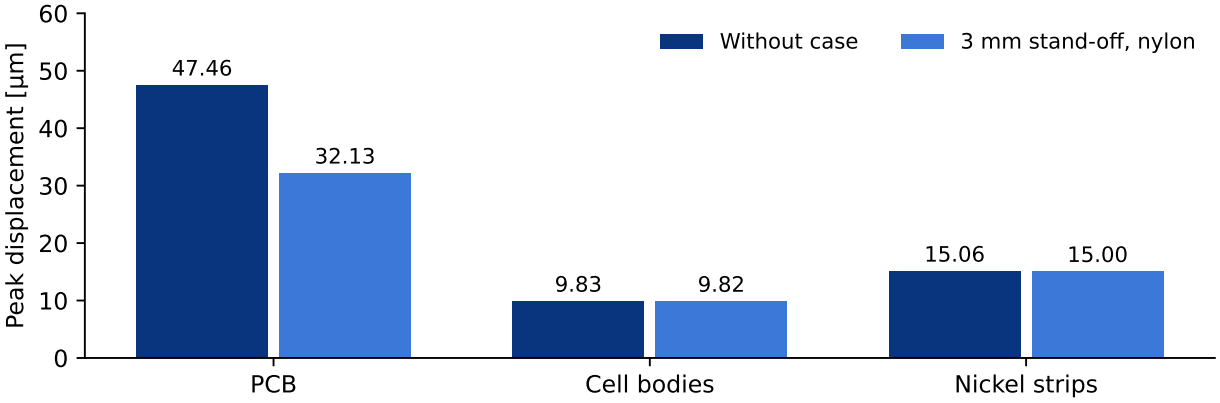


Figure 4: Peak displacement magnitude relative to the undeformed assembly after bolt locking. Each comparison uses the same mesh nodes on each part. Micrometres are used here to show the small changes at the cells and nickel strips.

The largest PCB displacement fell from 47.46 to 32.13 μm (32.3%). The maximum occurred at the same PCB node in both runs, and its motion was almost entirely normal to the board. The peaks on the cells and nickel changed by less than 0.4%. All 8,038 case-to-assembly contact faces remained open throughout tightening and their reported contact pressure was zero. The case therefore altered the board response through its modeled corner attachments. It did not brace the cell crowns during this loading. This differs from the lateral fixture, where the edge cell moved far enough to contact the case.

How the displacement and contact results were checked

The archived cased result uses a compressed displacement record that the legacy Mac reader misinterpreted. We recovered the translations from the original result file with ANSYS Data Processing Framework (DPF), without repeating the solve. On the bare control, DPF and the original reader agreed on every displacement component. On both runs the six pretension adjustments agreed with native MAPDL postprocessing within 10^{-11} m. We then compared 369,584 original solid-mesh nodes by ID and undeformed coordinates, including 217,704 on the PCB. Separately, contact-corner records and native face summaries agreed: some faces had a mixture of far and near corners, but none had a closed corner.

What remains to be established

This result concerns installation preload for one screw material and mounting height. It does not establish a reduction in vibration stress, cell motion under base excitation, or damage. The candidate flush model now starts from a checked open position and drives the screw heads towards the board relative to its flexible mounting points. A separate spring-and-mass verification reproduced the expected installation force and prestressed frequency; a converged flush installation and its stress and penetration limits still require verification.

The raised enclosure adds resonances below the first bare-board mode

We extracted undamped natural frequencies about the verified bolt-locked states, using the same mounting constraints and a 1–4000 Hz search. The assembled mass rose from 251.82 to 325.82 g. The calculation found 38 modes without the case and 49 with the raised nylon-screw case. In the 20–2000 Hz excitation band the counts were 13 and 20, respectively. The lowest three cased frequencies were 253, 289 and 339 Hz, all below the lowest bare frequency of 486 Hz.

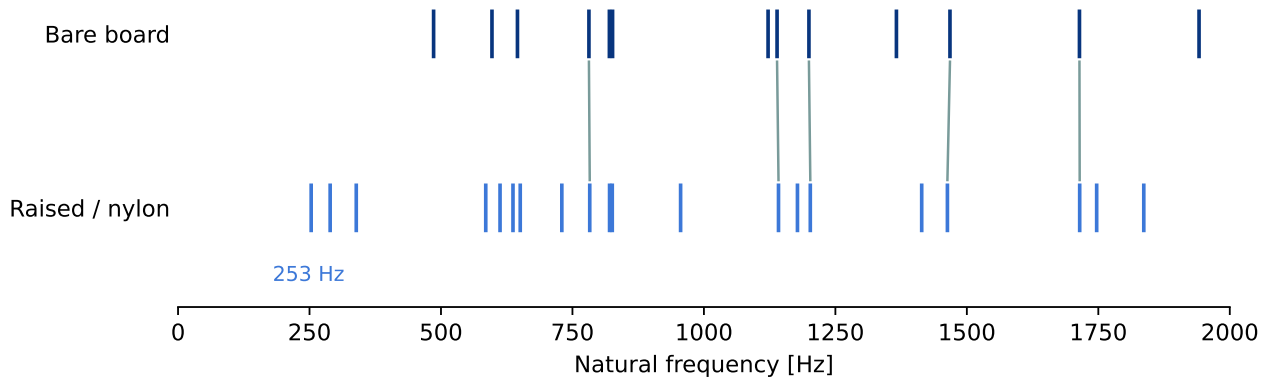


Figure 5: Each tick marks one native frequency in the excitation band. Lines connect five modes with a clear shape match (correlation above 0.94). Equal rank does not identify equal motion after adding the case.

The 253 Hz mode carried 17.9% of the assembled mass in X translation. In this shape the largest cell displacement was 2.8% of the peak on the PCB, while the peak on the case was 126% of the PCB peak. Those are amplitude ratios *within a mode*; the actual vibration displacement requires a forcing spectrum and damping. The 253 Hz mode had no close counterpart in the bare set: its highest shape correlation was 0.56. Conversely, the bare 781.35 Hz mode matched cased mode 9 at 782.88 Hz (correlation 0.943). This battery-motion mode shifted by only 0.20% even though its rank changed from fourth to ninth.

Effective translational mass, fraction of whole assembly	X	Y	Z
Bare, all 38 retained modes	87.2%	89.0%	89.6%
Raised nylon, all 49 retained modes	90.1%	91.5%	90.8%

How the shape comparison was made

Both modal meshes contain their assembly's static deformation. Before comparing vectors, we checked every original solid node against its own preload displacement and matched the 369,584 node IDs and their undeformed coordinates. The modal assurance criterion (MAC) is the squared, unweighted correlation of the global X, Y and Z eigenvector components at these nodes. It ranges from zero to one and is independent of the sign and normalization of a mode. It excludes the new case nodes so that both vectors describe the same structure. Modes with weak or competing matches remain ambiguous. Independently, all 38 bare frequencies and each printed per-mode translational mass fraction reproduced October. The new low frequencies establish additional ways the assembly can move. The next page examines forced displacement. The modal results alone give no stress amplitude.

The linear model predicts more PCB motion with the raised case

We applied the October base acceleration spectrum separately along X, Y and Z to the two bolt-locked assemblies. For the 3 mm stand-off with nylon screws, the independent calculation predicts greater out-of-plane PCB motion for all three excitation directions. Under X excitation the peak PCB response in Z rises from 23.37 to 145.55 μm RMS, a factor of 6.23. In the same calculation the largest cell response in X falls from 61.43 to 53.79 μm RMS (12.4%). The cell and board can respond differently to the same load. In particular, the 32% reduction in static board deflection during tightening did not carry over to this vibration metric.

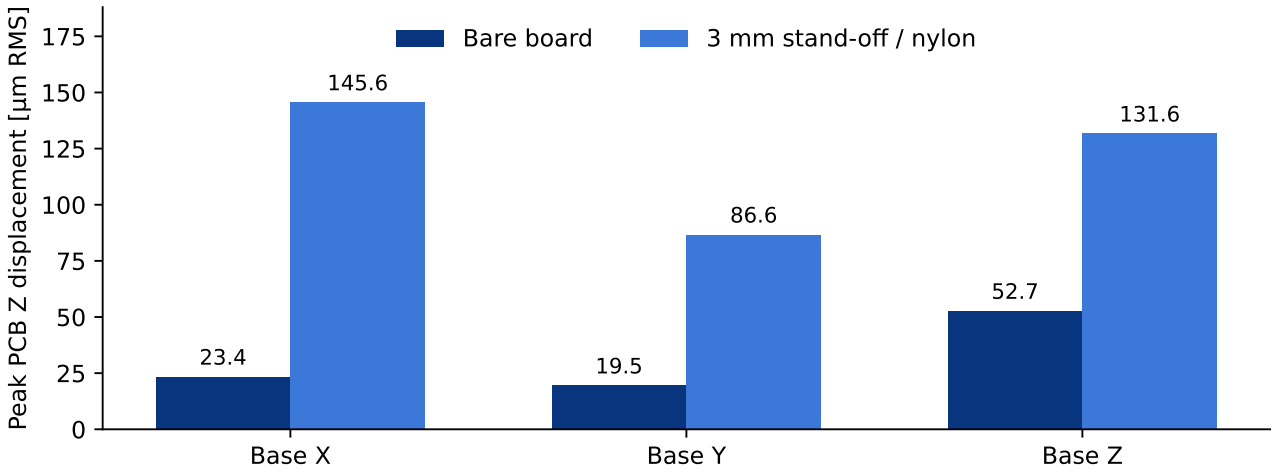


Figure 6: Each pair compares the largest board-normal RMS displacement anywhere on the PCB under the same single-axis base spectrum. Peaks can occur at different nodes. Values describe motion relative to the supports.

How we checked the response

The independent calculation uses each native mode's signed displacement field and directional participation factor. We integrate the modal frequency responses against the one-sided acceleration spectrum (20–2000 Hz, 14.14 g RMS, 2% modal damping), including cross terms between modes driven by the same base input. This preserves cancellation and reinforcement. Halving the quadrature spacing from 0.25 to 0.125 Hz changed the modal covariance by less than 6×10^{-8} in relative norm. For the bare assembly we compared all 18 peak directional displacements: three base directions, three response components, and both whole-assembly and PCB scopes. The worst difference from October's native ANSYS random-response output was 0.013%. A separate native repeat for base X reproduced all 17 October Mechanical maxima within 4.6×10^{-8} relative, including displacement, absolute acceleration, equivalent stress and the three normal elastic strains on the PCB. Its peak PCB equivalent stress was 90.325 MPa RMS.

This agreement checks the reconstruction against the bare reference. The raised results use the cased native mode shapes with an independent frequency integration; a cased native spectrum solve is still required for comparison. RMS is a statistical amplitude, not the largest displacement in a particular flight. The vibration calculation holds the installed contact state fixed. Separate maxima on the PCB, cells and case cannot be subtracted to determine whether an air gap closes. Clearance depends on the simultaneous motion of opposing surfaces along the local normal. These displacement results establish neither a stress margin nor damage. The following check tests the fixed-open assumption itself.

The 3 mm stand-off does not prevent dynamic cell contact in the model

The tab stand-off is not the air gap at the cells. The smallest sampled cell clearance on the tightened mesh was 0.143 mm. We paired nearby opposing surfaces and calculated their simultaneous relative motion along the local separation direction. Under base X the most exposed cell location had 0.145 mm clearance and 0.098 mm RMS relative motion. The gap there is only 1.47 times that RMS displacement. This ratio suggests contact is plausible, but it does not tell us when contact occurs.

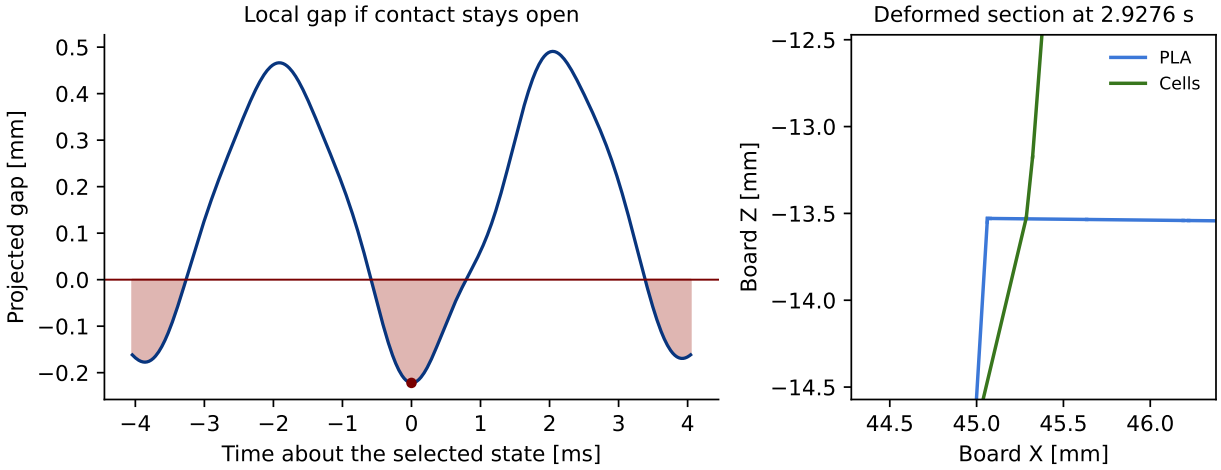


Figure 7: Raised case, nylon screws, October X-axis waveform. Left: first-order gap of the selected pair. A negative value indicates that the linear solution continues past projected contact. Right: actual-scale section of both triangulated surfaces displaced by the same signed modes. The curves are a diagnostic of the fixed-open model.

We independently integrated the nine-second October acceleration history and reconstructed the two skins at the instant of greatest projected approach for this pair, 2.9276 s into the waveform. At the tightened rest position the cell and case meshes had no intersections. In the selected deformed state, 895 triangle pairs intersected, and a separate signed-distance calculation placed 24 cell vertices more than 1 μm inside the PLA skin. The deepest sampled vertex was 0.214 mm inside. This is an impossible overlap allowed by the linear vibration solution. It identifies the need for a contact calculation; it is not a physical indentation or the time of first impact.

Excitation	Cell pair gap [mm]	Relative RMS [mm]	Intersecting cell/PLA pairs
Base X	0.145	0.098	895
Base Y	0.143	0.019	0
Base Z	0.236	0.112	0

Table 4: Each row selects the cell pair with the greatest RMS-to-gap ratio for that excitation, then checks the whole cell and case skins at one instant. A zero count at one instant does not establish separation throughout the record.

The surface search used both centroids and vertices of facing triangles within 1.2 mm in both directions, including the midside nodes of curved elements. Under base Z the nearest-line projection became negative, but the cell and case skins did not intersect at the selected instant. Tangential motion matters: a negative projected gap alone is insufficient evidence. Under base X both independent geometric checks found overlap. Until a nonlinear transient calculation or vibration test measures the response after closure, use the preceding fixed-contact RMS results as a comparison of the two linear models, not an installed flight stress, impact load or clearance margin.

The local flush installation stopped before the tabs seated

Four nylon screws drew the printed case toward a rigid board datum from an initial 3 mm stand-off. The existing washer clamps were tightened first. ANSYS saved 88 converged equilibria before the third load step failed at the next increment. The final converged screw-head travel was 2.8685 mm, leaving 0.1315 mm of the commanded travel. The solver never reached the sideways pull.

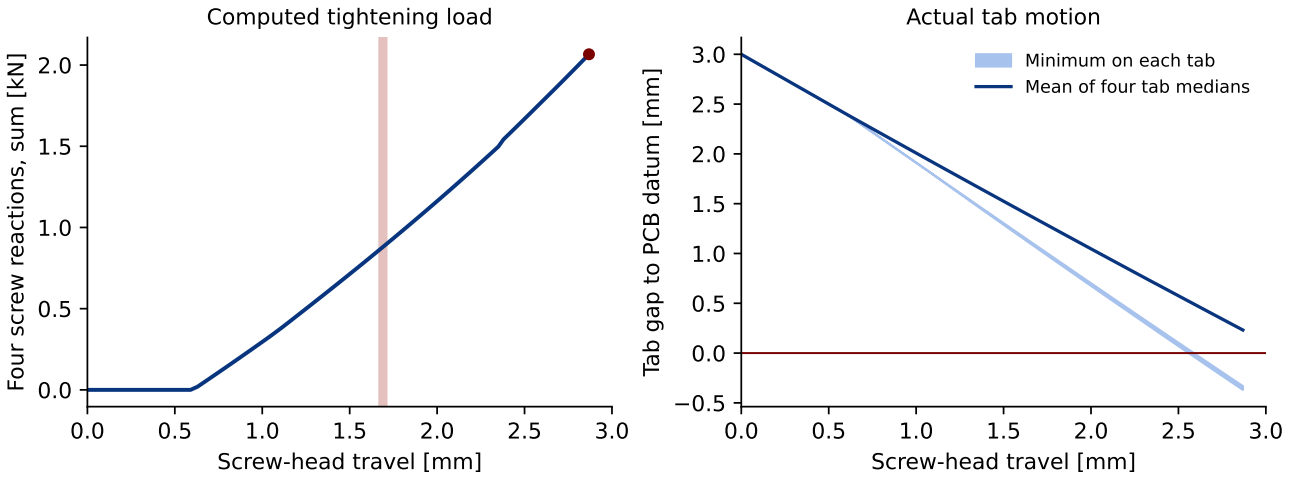


Figure 8: Left, the computed total screw reaction rose to 2.066 kN. The shaded interval marks the first exceedance of the 0.05 mm numerical penetration screen. Right, the tab surfaces tilted during drawdown. At the last equilibrium every tab crossed the nominal PCB plane at some points and remained clear at others.

The first new contact was with the fixed surfaces representing the neighboring cells and washers at 0.59–0.63 mm screw travel. The moving edge cell and its two nickel strips remained clear of the PLA throughout this partial path. Loaded-corner penetration first exceeded 0.05 mm at 1.67–1.71 mm travel and reached 0.119 mm. The largest departure from force balance was 0.030 N. This force balance shows that the reported reactions support the forces in the numerical model. The penetration and tab geometry limit their physical interpretation.

Final saved equilibrium	Tab 1	Tab 2	Tab 3	Tab 4
Screw-head reaction [N]	484	517	530	536
Minimum gap to board datum [mm]	−0.332	−0.357	−0.366	−0.369
Median sampled gap [mm]	0.248	0.227	0.236	0.214

In this local fixture the seating springs moved with the screw grips, allowing a tilted tab to cross the board plane. The peak computed PLA equivalent stress reached 233 MPa under the elastic print model. The physical installation force, failure sequence, and full-flush side-pull response remain unresolved. The whole-board installation model instead places contact directly between each tab and the flexible PCB surface, so the model can develop a normal reaction at the bearing land.

Contact deflects the raised case along the middle cell

The fixed-open calculation on page 10 placed parts of all three cell surfaces inside the enclosure at the selected X-waveform sample. Along the middle cell, the affected PLA facets extended from 25.2 to 95.6 mm in board Y. We therefore retained the whole length of this contact region in two native submodels. Their outer boundaries lay 9 or 15 mm from the middle-cell contact line. Each retained the original quadratic elements and directional PLA properties.

For each domain we imposed the tightened board displacements, then applied the signed motion at 2.9276 s to the two local cell bodies and the outer cut faces of the PLA. We increased the imposed motion from zero to its full value and unloaded in a series of static equilibria. With cell contact disabled, the free PLA nodes followed the parent solution to within 1 μm. With frictionless contact active, the largest change in their motion was 249 μm in the smaller domain and 244 μm in the larger one.

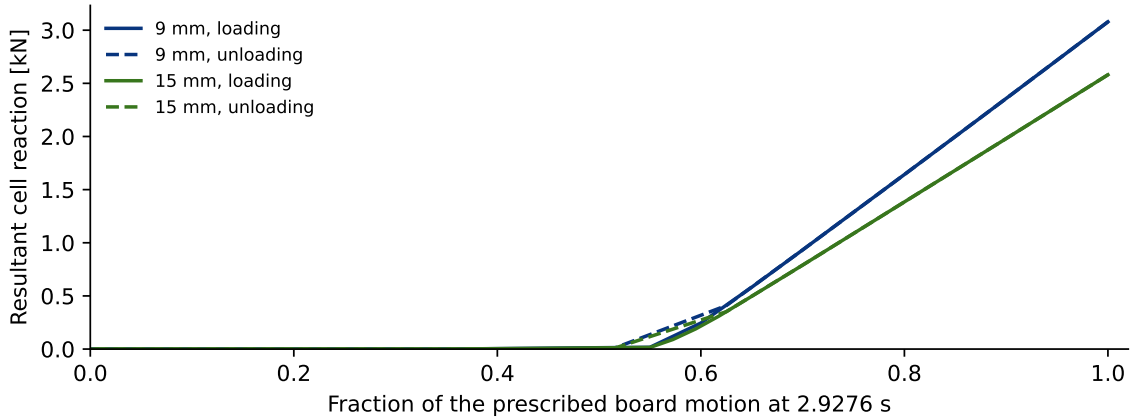


Figure 9: Each curve gives the force needed to impose this particular displacement path. The larger domain leaves a wider section of PLA free to flex. The peak reaction fell by 16.2%. The local models hold the cell motion fixed as the case pushes back.

At the fully imposed displacement	9 mm domain	15 mm domain
Resultant cell reaction [kN]	3.079	2.581
Active contact faces	24	24
Active faces touching the artificial cut	0	0
Maximum sampled skin overlap, contact disabled [mm]	0.2165	0.2164
Maximum sampled skin overlap, contact enabled [mm]	0.0424	0.0445

The independent geometry check measured signed distances on vertices and facet centres inside a common 8 mm band. Some tessellated surfaces still intersected even though the largest loaded-corner penetration recorded by ANSYS was below 0.5 μm at this state. Changing the cut size changed the force appreciably despite whole-model force imbalances below 0.004 N. These results establish local deflection under imposed motion. A flight impact load requires the cells and the surrounding board to respond to contact together.

A separate search near the beginning of the same X waveform found a clear middle-cell state at 85.3424 ms and intersecting skins at 85.3577 ms. The search sampled at 8192 Hz and refined that interval at 65,536 Hz. Contacts between earlier survey samples or elsewhere on the assembly remain possible. This interval sets the time window for the transient contact comparison.